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ORIGINAL ARTICLE

Evaluation of a novel large language model (LLM)-powered chatbot for oral boards scenarios

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Abstract

Purpose While previous studies have demonstrated that generative artificial intelligence (AI) can pass medical licensing exams, AI's role as an examiner in complex, interactive assessments remains unknown. AI-powered chatbots could serve as educational tools to simulate oral examination dialogues. Here, we present initial validity evidence for an AI-powered chatbot designed for general surgery residents to prepare for the American Board of Surgery (ABS) Certifying Exam (CE).

Methods We developed a chatbot using GPT-4 to simulate oral board scenarios. Scenarios were completed by general surgery residents from six different institutions. Two experienced surgeons evaluated the chatbot across five domains: inappropriate content, missing content, likelihood of harm, extent of harm, and hallucinations. We measured inter-rater reliability to determine evaluation consistency.

Results Seventeen residents completed a total of 20 scenarios. Commonly tested topics included small bowel obstruction (30%), diverticulitis (20%), and breast disease (15%). Based on two independent reviewers, evaluation revealed 11–25% of chatbot simulations had no errors and an additional 11%–35% contained errors of minimal clinical significance. The chatbot limitations included incorrect management advice and critical omissions of information.

Conclusions This study demonstrates the potential of an AI-powered chatbot in enhancing surgical education through oral board simulations. Despite challenges in accuracy and safety, the chatbot offers a novel approach to medical education, underscoring the need for further refinement and standardized evaluation frameworks. Incorporating domain-specific knowledge and expert insights is crucial for improving the efficacy of AI tools in medical education.

Keywords Artificial intelligence (AI) · GPT · Oral boards · Graduate medical education (GME) · Simulation

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Introduction

General surgery trainees who seek board certification are required to pass the American Board of Surgery (ABS) Certifying Exam (CE) and Qualifying Exam (QE). The ABS CE is an oral examination that evaluates knowledge and clinical reasoning skills concerning various surgical conditions, as outlined by the ABS [1]. The candidates may only attempt the exam once per academic year, making a first-time pass helpful for a successful career launch. The surgical trainees face considerable pressure from multiple stakeholders to pass their ABS CE. The accreditation of residency programs depends on board exam pass rates [2], future employers often base privileges on board certification [3, 4], and the public generally assumes that board-certified surgeons deliver better outcomes—an assumption increasingly supported by a body of research [5–7]. Given the high stakes and the exam's unique format, general surgery trainees often find the experience both stressful and intimidating. Implementing effective strategies to support trainees in preparation for the ABS CE is, therefore, essential for residency programs.

Despite the significance of the CE, the preparation is challenging because the ABS does not disclose specific topics or the evaluation framework. The ABS indicates that the content of the CE "generally, though not exclusively" aligns with the Surgical Council on Resident Education (SCORE) curriculum [1]. Consequently, general surgery residents typically prepare through a combination of direct patient care and targeted didactics that follow the SCORE curriculum [8, 9]. The chief residents commonly use study materials, such as textbooks and podcast series [10, 11] and while many commercial online review courses are available, they are expensive and often lack efficacy or outcomes data [12]. Many general surgery residency programs implement mock oral examinations (MOEs) to prepare trainees and assess CE readiness. Research shows a positive correlation between MOE participation and first-time oral board pass rates [2, 13, 14], highlighting the value of MOEs in identifying individuals at risk [15] and curricular gaps [16, 17]. However, MOE implementation is limited by high operational costs, logistical challenges, limited availability of faculty examiners, and a scarcity of standardized materials, complicating their sustainability [18]. Given how resource intensive in-person MOEs can be to create, organize, and execute, they can be infrequent and geographically limited. Moreover, because MOEs are typically conducted in front of peers, they may discourage participation among more reserved trainees due to their public nature [19, 20].

Recent advances in AI have led to the development of large language models (LLMs), which offer potential solutions to these challenges. LLMs are algorithms trained on large data sets, enabling them to recognize, translate, predict, or generate text, giving them the sobriquet of "generative AI"[21]. These models have been employed to generate ABS In-Training Examination (ABSITE) questions [22], create diagnostic and patient management vignettes [23], and simulate virtual patients for practicing history taking skills [24]. Furthermore, models such as Chat Generative Pre-trained Transformers (ChatGPT) have popularized the practical use of LLMs through chatbot interfaces, mimicking conversational interfaces [22]. Given that the ABS CE is a conversational exam, we hypothesized that LLMs could replicate the oral board experience. Initial investigations into LLMs in medical education have primarily assessed their medical knowledge, particularly their capability to succeed in written board examinations consisting of multiple choice questions [25–29]. More recently, the studies have begun to look into the models' complex reasoning abilities via free response questions [29–31]. Despite their successes, the accuracy and safety of employing such technologies as virtual patients in preparations for complex and nuanced assessments like the ABS CE remain unexplored. This gap highlights the need to investigate LLMs' capabilities in the nuanced contexts of graduate medical education (GME), especially in roles beyond traditional examination settings.

This study aims to leverage advancements in AI and LLMs to develop a specialized tool tailored for general surgery residents preparing for the ABS CE. By focusing on the development and utilization of a chatbot that harnesses an LLM, our primary objectives were to evaluate the chatbot's (1) accuracy in simulating patient interactions and (2) safety in providing medically sound feedback. Through rigorous assessment, we intend to determine the model's efficacy and reliability as an educational tool, thereby addressing a current gap in the application of AI technologies within GME. Ultimately, our goal is to establish the model's potential as a viable and innovative educational resource for enhancing the preparation of surgical residents for their ABS CE.

Methods

Ethical consideration

This project was classified as exempt research by the Columbia University Institutional Review Board and did not require a full review under Protocol AAAV2136.

Chatbot development

We developed a custom chatbot leveraging the GPT-4 LLM from OpenAI to simulate the dynamics of general surgery oral board scenarios (OpenAI, San Francisco, CA). The version of GPT-4 used for this study was based on data available up to September 2021.

The development process involved three main stages (Appendix A):

Interface creation: Vercel (https://vercel.com/) was utilized to craft a user-friendly interface for interaction with the chatbot. Using this, we designed a static website and user interface (Vercel, San Francisco, CA). There was no cost associated with using the free tiers of Vercel or Supabase for this project.

Database integration: a supabase-based database was established to support the chatbot's operations, developed as a PostgreSQL database. This allowed us to communicate with the GPT-4 model from Open AI (Supabase, San Francisco, CA).

AI integration: an OpenAI application programming interface (API) key facilitated communication between the database (Supabase) with the GPT-4 model, enabling the chatbot to generate realistic case scenarios. The cost of using the GPT-4 API is based on the number of tokens, which are pieces of words or whole words, used in both input prompts and output responses. 1000 tokens cost around $0.03 and are equal to roughly 750 words.

To ensure the accuracy and relevance of the generated responses, we employed a systematic prompting technique known as prompt engineering. Prompt engineering, which refers to the systematic design and optimization of input prompts, helps guide the responses of language models, ensuring the accuracy, relevance, and coherence of the generated output for specific use cases [32]. This critical process enhances the model's ability to produce accurate responses and reduces the likelihood of generating incorrect information [33]. By applying fundamental principles of prompt engineering, along with knowledge of the oral boards provided by the ABS website [1] and Reviewer 1 (WDW), we designed a prompt to simulate the experience of taking the ABS CE, instructing the chatbot to guide users through cases in the style of general surgery oral boards (Appendix B). The chatbot was then prompted to create cases at random, using a single topic from the list of entrustable professional activities (EPAs) for general surgery [34]. Each case started with a patient scenario that included patient demographic data (age and sex), chief complaint, and care setting. Individual users were instructed to interact with the model as they would during a MOE, but were allowed to respond freely without specific guidance on how to structure their answers. Using natural language conversation, the chatbot led users through aspects of history and physical, vitals, laboratory testing, imaging, diagnosis and management. If operative management was required for the case, the chatbot prompted the user to detail the preoperative work up, main operative steps, post-operative management, and management of common complications. At the end of the case, the chatbot prompted users to reflect on their performance and then provided detailed feedback using assessment criteria obtained from the ABS [1]. Lastly, we prompted the chatbot to provide a summary of the case including work-up, diagnosis, and management of the patient along with the postoperative complication. An example of a case is provided in the supplementary documents (Appendix C). To ensure the accuracy and relevance of the generated responses, we piloted the introductory prompt using various EPA topics. We carefully assessed any deviations or errors from the intended instructions. Identified errors led us to adjust the introductory prompt's language and this process was iteratively repeated until the outputs were following the prompt instructions (e.g. only providing cases within EPA topics provided, following logical progression of the case, formatting vitals and labs as described, only offering information to the user that was directly asked for).

Participants

We enlisted members of the Collaboration of Surgical Education Fellows (CoSEF), a network of General Surgery residents engaged in surgical education scholarship, to recruit residents to participate in this study [35]. CoSEF members recruited participants at their institutions via email, which included only general surgery residents. The emails contained all research study materials and detailed instructions for participation. We gave residents at each participating institution an institution-specific username and password to access the chatbot. Participants voluntarily and anonymously engaged in oral board scenario simulations. Users were instructed to complete individual case scenarios in a private window of any web browser, to avoid the model learning from prior cases. Post-simulation, they were invited to share optional demographic data, including postgraduate year (PGY) level and gender. Institutional data for each participant was intuited from the login that they used.

Evaluation rubric development

To evaluate the chatbot, we modified a rubric previously used to gather validity evidence for LLM responses to patient-posed questions [36]. The responses generated by the chatbot throughout the case scenario, as well as the feedback section to the user on their performance, were subject to evaluation. The rubric assesses the chatbot's responses in five key domains: (1) inappropriate content, (2) missing content, (3) likelihood of harm, (4) extent of harm, and (5) hallucinations (Appendix D).

We had three objectives in our evaluation of the chatbot. First, we aimed to ascertain the completeness and accuracy of the responses generated by the chatbot. This entailed a detailed evaluation of the presence of any missing or incorrect content within the chatbot's responses. The reviewers were tasked with determining whether the information relayed by the chatbot included erroneous elements or omitted essential details pertinent to the case scenarios. In instances where missing or incorrect content was identified, reviewers were required to further categorize the clinical significance of these discrepancies, distinguishing between those of great significance and those of minimal clinical significance.

Second, we aimed to ascertain the potential severity and likelihood of harm associated with the chatbot's generated information, should users implement this information in a clinical setting. The reviewers were tasked with evaluating the implications of the chatbot's outputs, assuming these were accepted as accurate by the users and acted upon clinically. The assessment focused on determining the potential for physical or mental harm, categorizing the likelihood of such harm on a scale ranging from low, medium, or high. Furthermore, the evaluation of harm severity employed the Agency for Health Care Research and Quality (AHRQ) common formats as a reference point, allowing for the classification of harm into categories spanning from no harm, to mild to moderate harm, to severe harm or death [36, 37]**.**

Lastly, we wanted to understand whether the chatbot was generating hallucinations, defined as instances where the chatbot generated nonexistent, false, or misleading information with confidence [38]. The reviewers were instructed to identify hallucinations and then asked to categorize these hallucinations based on their clinical significance. To assist with this, we provided examples from our pretesting phase, illustrating both major and minimal clinical significance. The major hallucinations were those that could significantly impact the case scenario or lead to clinical errors or patient harm if believed to be true. The minimal hallucinations were those unlikely to affect the case scenario or patient outcomes. The reviewers used these criteria to assess and categorize the hallucinations.

In addition to the quantitative aspects of the rubric, each reviewer was given the opportunity to provide qualitative comments within each of the domains. In the analysis of transcripts, we defined perfect quality transcripts as those that contained no errors and good quality transcripts as those that contained errors of minimal clinical significance.

Reviewer selection

Two highly qualified, board-certified general surgery faculty, with extensive experience in oral boards, were selected to grade chatbot responses using the rubric described above. Reviewer 1 (WDW) is a Clinical Professor of Surgery and has over a decade of experience developing and running a national oral board review course for general surgery residents. Reviewer 2 (CT) is a Professor of Surgery, elected American College of Surgeons (ACS) Master Surgeon Educator, past President of the ACS, prior Chair of the ABS, and retired ABS oral board examiner.

A standardized set of instructions was provided to orient the reviewers. These instructions included the logistics of transcript formatting to ensure reviewers could distinguish between text from the chatbot and the user. It also included details on the prompt structure of a case scenario and the different parts of the scenario they would be evaluating. We also provided comprehensive guidance on the contents and use of the rubric, including a detailed explanation of hallucinations and an example identified during our pretesting. We further described that the chatbot may generate conclusions or clinical findings that were not present in the input data or that it can fabricate connections that are not medically sound. For example, when practicing a case of sigmoid volvulus, the bot asked the user to perform a sigmoid resection. After the user asked the bot for help, it gave us the correct steps of the operation but referenced "removal of a tumor" which may be a part of some sigmoid resections but was nonsensical in the context of that case, especially given the preoperative colonoscopy negative for any masses. After reviewing the instructions and the rubric, the reviewers were provided with a sample case scenario to practice evaluation. Any questions that arose during this process were discussed with the research team before starting the formal evaluation of the standardized transcripts.

Analytic approach

We conducted descriptive statistical analyses on the demographic data provided by participants upon completion of the case scenarios. Similarly, we subjected the content of the case scenarios to descriptive statistical evaluation. To visually represent the assessment outcomes, we generated individual heat maps for each reviewer, showcasing the analysis of each transcript across the designated rubric domains. To assess the consistency of ratings between reviewers, we calculated inter-rater reliability for each domain of the rubric using Cohen's Kappa. Furthermore, we calculated the overall reliability across entire transcripts using a weighted Cohen's Kappa. Given the novelty of the our assessment tool, when interpreting the calculated Cohen's Kappa we relied on ranges and interpretations outlined by Cohen in his initial analysis [39, 40]

Data and code

Data analysis was conducted by a data analyst with statistics background (SMN) and separately confirmed with Advanced Data Analysis by GPT-4. All analyses were performed using Python 3.10 (Python Software Foundation, Wilmington, DE). Our code is available here: [https://github.com/tsathe/surgbot-data-analysis].

Results

Participant demographics

Seventeen general surgery residents across six training programs participated in the study by completing at least one chatbot simulation (Table 1). Of the 17 participants, 15 provided demographic data. Of those participants, 8 (53%) were male and 7 (47%) were female. The cases were most commonly completed by PGY3 residents ($n=6$, 35%).

Characteristics of case scenarios

The residents completed a total of 20 oral board case scenarios. Commonly tested topics included small bowel obstruction ($n=6$, 30%), diverticulitis ($n=4$, 20%), and breast disease ($n=3$, 15%) (Table 2). In 18 cases (80%), users experienced a patient complication during the case. For cases requiring operative management, complications included preoperative, intra-operative, and post-operative issues. In cases where operative management was not required, the complications arose during the patient's hospital course (Table 3). In two (10%) cases, no complication was given to the user during the case scenarios.

Individual raters review

All 20 transcripts were reviewed by Reviewer 1. Twelve (60%) had no inappropriate content while three (15%) and five (25%) had inappropriate content of minimal and high clinical significance, respectively. Eleven transcripts (55%) had no missing content while five (25%) and four (20%) had missing content of minimal and high clinical significance, respectively. Sixteen (80%) had low likelihood of harm while three (15%) and one (5%) had medium and high likelihood of harm, respectively. Fifteen (75%) demonstrated no harm while four (20%) and one (5%) demonstrated mild or moderate and severe harm or death, respectively. Only one transcript (5%) exhibited a hallucination of high clinical significance. In this case, the chatbot described the CT scan as showing a transition point in the proximal ileum with dilation. However, intra-operatively, a hard mass was found involving the terminal ileum and cecum. Five transcripts (25%) exhibited the highest score possible on our rubric (Fig. 1).

Nineteen transcripts were reviewed by Reviewer 2. Reviewer 2 recused themselves from evaluating Transcript 19, citing not having enough clinical expertise with

Table 1 Demographic data of resident participants completing oral case scenarios

Characteristic	Frequency (%), $N=17$
Gender	
Male	8 (47%)
Female	7 (41%)
Unknown	2 (12%)
Post-graduate year (PGY)	
1	2 (12%)
2	0 (0%)
3	6 (35%)
4	3 (18%)
5	4 (23%)
Unknown	2 (12%)

Table 2 Type and frequency of case topic generated by the AI-model, based on reference list of EPA topics, for each oral board scenario

Case topic	Frequency (%), $N=20$
Small bowel obstruction	6 (30%)
Diverticulitis	4 (20%)
Breast disease (Benign or Malignant)	3 (15%)
Inguinal hernia	2 (10%)
Appendicitis	1 (5%)
Pancreatitis	1 (5%)
GYN pathology	1 (5%)
Trauma	1 (5%)
Ascites	1 (5%)

Table 3 Type and frequency of complications assessed during oral case scenarios

Complication (intra-op, post-op)	Frequency (%), $N=20$
None	2 (10%)
Small bowel obstruction	2 (10%)
Post-op seroma	2 (10%)
Post-op intra abdominal abscess	2 (10%)
Worsening diverticulitis, post-op pulmonary embolism	2 (10%)
Post-op fever	2 (10%)
Post-op stoma necrosis	1 (5%)
Necrotizing and infected pancreatitis	1 (5%)
Anastomotic leak	1 (5%)
Abdominal compartment syndrome	1 (5%)
Intraoperative bleeding	1 (5%)
Rib fractures with effusion and atelectasis	1 (5%)
Intraoperative bowel ischemia requiring resection	1 (5%)
Post-op hematoma	1 (5%)

Transcript Number	Inappropriate Content	Missing Content	Likelihood of Harm	Extent of Harm	Hallucinations
1	Great clinical significance	Great clinical significance	Low	Mild or Moderate Harm	No
2	No	Little clinical significance	Low	No Harm	No
3	No	Little clinical significance	Low	No Harm	No
4	No	No	Low	No Harm	No
5	Great clinical significance	No	Low	No Harm	No
6	No	No	Low	No Harm	No
7	Little clinical significance	No	Medium	No Harm	No
8	No	No	Low	No Harm	Great clinical significance
9	No	No	Low	No Harm	No
10	Little clinical significance	Great clinical significance	Low	No Harm	No
11	No	Little clinical significance	Low	No Harm	No
12	Little clinical significance	No	Low	No Harm	No
13	No	No	Low	No Harm	No
14	Great clinical significance	Great clinical significance	Medium	Mild or Moderate Harm	No
15	No	Great clinical significance	High	Severe Harm or Death	No
16	No	Little clinical significance	Low	No Harm	No
17	No	No	Low	No Harm	No
18	No	Little clinical significance	Low	Mild or Moderate Harm	No
19	Great clinical significance	No	Low	No Harm	No
20	Great clinical significance	No	Medium	Mild or Moderate Harm	No

Fig. 1 Transcript error heatmap for general surgery oral boards simulation cases—Reviewer 1 (WDW)

the scenario (medical management of ascites) to evaluate. Twelve (63%) had no inappropriate content while four (21%) and three (16%) had inappropriate content of minimal and high clinical significance, respectively. Two transcripts (10.5%) had no missing content while two (10.5%) and fifteen (79%) had missing content of minimal and high clinical significance, respectively. Seven (37%) had low likelihood of harm while two (10%) and ten (53%) had medium and high likelihood of harm, respectively. Six (32%) demonstrated no harm while four (21%) and nine (47%) demonstrated mild to moderate and severe harm to death, respectively. Three transcripts (16%) exhibited a hallucination with minimal clinical significance. Two transcripts (10.5%) exhibited the highest score possible on our rubric (Fig. 2).

Reviewer 1 found 25% of transcripts to be perfect quality (with no errors) and an additional 35% to be good quality (with only errors of minimal clinical significance). Reviewer 2 found 10.5% of transcripts to be perfect quality and an additional 10.5% of transcripts to be good quality.

Interrater reliability

We calculated each reviewer's total score for each transcript and interrater reliability yielded none to slight agreement between total scores (Weighted Cohen's Kappa = 0.12). When evaluating each individual rubric domain, we found varying levels of agreement.

Inappropriate content

In the domain of appropriate content, interrater reliability yielded fair agreement, with a Cohen's Kappa of 0.31. In transcripts where inappropriate content was noted, reviewer comments were reviewed. The majority of inappropriate content was indicated when reviewing the feedback section

Transcript Number	Inappropriate Content	Missing Content	Likelihood of Harm	Extent of Harm	Hallucinations
1	Little clinical significance	Great clinical significance	Low	No Harm	No
2	Little clinical significance	Great clinical significance	Medium	Mild or Moderate Harm	Little clinical signifcance
3	No	Great clinical significance	High	Severe Harm or Death	No
4	No	Great clinical significance	High	Severe Harm or Death	Little clinical signficance
5	Great clinical significance	Great clinical significance	High	Severe Harm or Death	No
6	No	Little clinical significance	Low	No Harm	No
7	No	Great clinical significance	Low	Mild or Moderate Harm	No
8	No	Great clinical significance	Medium	Severe Harm or Death	No
9	No	Great clinical significance	High	Severe Harm or Death	Little clinical signficance
10	Little clinical significance	Great clinical significance	High	Severe Harm or Death	No
11	No	Great clinical significance	High	Severe Harm or Death	No
12	No	No	Low	No Harm	No
13	No	Great clinical significance	High	Severe Harm or Death	No
14	Great clinical significance	Great clinical significance	Low	No Harm	No
15	Little clinical significance	Great clinical significance	High	Severe Harm or Death	No
16	No	No	Low	No Harm	No
17	No	Little clinical significance	Low	No Harm	No
18	Great clinical significance	Great clinical significance	High	Mild or Moderate Harm	No
19	Declined to answer	Declined to answer	Declined to answer	Declined to answer	Declined to answer
20	No	Great clinical significance	High	Mild or Moderate Harm	No

Fig. 2 Transcript error heatmap for general surgery oral boards simulation cases—Reviewer 2 (CT)

produced by the bot. Inappropriate content included domains such as inappropriate management (e.g., bot gave inappropriate follow up recommendations or inappropriate feedback on timing of antibiotics or when to obtain a CT scan), inappropriate differential diagnosis (e.g., bot said Budd-Chiari was on the differential despite labs not being consistent), inappropriate surgical technique description (e.g., bot reviewed steps of open inguinal hernia repair and indicated inappropriate approach to mesh fixation), and inappropriate corrective feedback to the user (e.g., user did not act on a symptomatic seroma and bot did not correct them).

Missing content

In the domain of missing content, interrater reliability yielded none to slight agreement, with a Cohen's Kappa of 0.02. For Reviewer 2, more transcripts were noted to have missing content when compared to Reviewer 1. For Reviewer 1, the comments for missing content indicated diagnostic and follow-up omissions (e.g., lack of genetic testing when applicable on family history, lack of pregnancy test prior to CT scan), missing preoperative preparation (e.g., lack of re-evaluation of patient prior to OR take-back), surgical approach omissions (e.g., not correcting user when proceeding with mini laparotomy when laparoscopy would be indicated, omitted peritoneal washings when indicated), and critical care management omissions (e.g., not providing specifics of intravenous fluid resuscitation for unstable patients). For Reviewer 2, the comments for missing content mostly focused on components lacking from the comprehensive case summary at the conclusion of the case, indicating the chatbot generated summary should review all possible aspects of care for users to consider in future cases. The reviewer comments included lack of adequate summary of comprehensive patient history (e.g., family, past medical, past surgical), diagnostic testing recommendations (e.g., wide and comprehensive range of diagnostic tests that could influence patient care decisions and management), and detailed management plans (e.g., specifics on fluid resuscitation, specifics on nasogastric tube outputs and rates affecting management, getting previous operative reports) to consider in future cases.

Likelihood of harm & extent of harm

In the domain of likelihood of harm and extent of harm, interrater reliability yielded no agreement and none to slight agreement with Cohen's Kappa of − 0.03 and 0.09, respectively. Similar to the prior category of missing content, Reviewer 2 noted more transcripts to have a high likelihood of harm and larger extent of harm when compared to Reviewer 1. Notes indicated that Reviewer 2 believed that a lack of comprehensive feedback and case summary produced by the chatbot for the user would result in a high likelihood of harm and severe extent of harm in most cases.

Hallucinations

In the domain of hallucinations, interrater reliability yielded no agreement with Cohen's Kappa of − 0.04. A total of four hallucinations were identified between both reviewers, and it was noted that 3 (75%) occurred during case presentations, while 1 (25%) was noted in the feedback and case summary section. Specifically, during the case presentations, the chatbot generated findings or conclusions not supported by the clinical data. For instance, one hallucination involved a report of a hard mass found intra-operatively in the terminal ileum (TI), despite CT scans indicating a transition point in the proximal ileum. Similarly, another scenario involved a report of a fecalith found intra-operatively in the TI, with incongruent CT scan findings and patient factors in the setting of a small bowel obstruction. The last hallucination involved a postoperative day three small bowel anastomosis leak, which the bot incorrectly attributed to an anastomotic stricture—an unlikely occurrence without a technical error—later citing the patient's underlying Crohn's disease as a risk factor. These instances are categorized as hallucinations because the chatbot introduced fabricated clinical details or incorrect reasoning that were not derived from the input data. Conversely, the sole hallucination identified within the feedback section involved the assertion that small bowel obstructions are a common postoperative complication, supported by an erroneously cited incidence of 3%.

Discussion

Our study explores the innovative integration of an LLM-based chatbot designed to simulate the ABS CE. In our reviewers' evaluations, the model was able to take a user through a complex patient scenario regarding diagnosis, work up, non-operative and operative management, and common complications, achieving perfect quality (no errors) in 11%–25% of cases and good quality (with only errors of minimal clinical significance) in an additional 11%–35% of cases. While the custom model demonstrated commendable promise with varying degrees of accuracy, our study also revealed its current inadequacies for independent use without oversight. Notably, there were cases where the model produced inaccurate information regarding patient management, however, the more significant inadequacies noted by reviewers were omissions of critical information within the scenario or not providing critical feedback to the user on mistakes they made.

Consistent with existing research, our study highlights the potential of AI in enhancing surgical education by providing novel, personalized, and cost-effective tools for learners. Traditional simulation models can often be constrained by financial costs, the availability of faculty experts, and the significant time required to develop and implement content. Moreover, these models can lack the personalized feedback and custom learning plans necessary to meet the unique needs of each learner. Generative AI has been shown to overcome these limitations by generating novel appearing content [41, 42], creating virtual simulations [24], and providing evaluation of operative skills [43]. Additionally, unlike traditional computer simulation models, generative AI offers learners a more dynamic, realistic, and challenging simulation experience [44, 45]. This technology also enhances trainee evaluations by providing personalized assessments, real-time feedback, and customized learning plans [44, 46], all while reducing time and costs [44]. Additionally, AI can generate a wide range of simulated patient scenarios, enabling trainees to practice clinical skills in a low-stakes environment [42, 47]. While traditional simulations with computers or humans offer a high degree of realism, AI-driven simulations provide a cost-effective alternative that democratizes access to high-quality simulations [45].

Aligning with existing research, our study showed that despite its benefits, utilization of this technology provides significant challenges. The quality of the AI-generated content has been shown to oftentimes be inaccurate or create hallucinations, so educators must meticulously assess their models to ensure accuracy, relevance, and safety [45]. The task of estimating the uncertainty of models' output is an important problem and has seen considerable development within general machine learning and deep learning domains, with few focusing on LLMs [48, 49]. Therefore, the researchers have instead focused on measures shown to improve the accuracy of these models and decrease hallucinations, which include effective prompt engineering and utilization of iterative feedback loops [50, 51]. Additionally, incorporating domain-specific data, utilizing retrieval augmented generation (RAG) and fine-tuning, can increase accuracy. RAG involves adding domain-specific knowledge to the initial prompt, while fine tuning involves retraining the model parameters with a more topically relevant or accurate dataset [52, 53]. Both of these processes can increase model performance and reliability, especially when being used for specific tasks.

Furthermore, in line with the challenges of ensuring model accuracy, our study brings to light the critical need for standardized rubrics with validity evidence for the assessment of accuracy and safety of AI models in medical education. Currently, there are well-established rubrics for evaluating various aspects of simulation-based learning. For example, the Michigan Standard Simulation Experience Scale (MiSSES) provides a comprehensive, yet tailored, instrument for capturing subjective measures in surgical simulation [54]. Similarly, the Objective Structured Assessment of Technical Skills (OSATS) is widely used to assess technical skills across various surgical simulations, while the Global Operative Assessment of Laparoscopic Skills (GOALS) is specifically designed for evaluating laparoscopic technical skills [55, 56]. These validated tools for assessing different aspects of surgical simulation training can serve as a foundation for developing evaluation methods tailored to AI-powered simulators. The absence of a shared evaluation framework for the quality of AI-generated education tools poses a challenge in ensuring alignment with medical standards and educational goals. Without standardized evaluation metrics and a clear understanding of AI model capabilities, discrepancies can arise among human evaluators. These differences may stem from inconsistent interpretations of the evaluation tools or varying expectations of what the models should be capable of, rather than of the models' output.

Implications for surgical education

The integration of AI in GME, particularly within surgical training, marks a departure from traditional teaching methodologies. Currently, the prevailing education model heavily relies on the apprenticeship model, necessitating regular exposure and repetition of direct patient care encounters. The restrictive work hours and increase in nonclinical administrative duties have contributed to the decrease in these encounters and operative experience in residency [57]. While simulation has emerged as a platform for residents to get these experiences in controlled environments, several impediments persist in the effective utilization of this technology, including instructor availability, protected time for education, financial burdens, and lack of immediate feedback [58–60]. AI has already been integrated into teaching procedural skills to surgical residents through the use of virtual reality and augmented reality, and studies have begun to use AI to provide real-time feedback to trainees performing surgical tasks [43].

Despite these advancements, clinical simulations that focus on preparing trainees for oral exams, have not been extensively studied. Preparation for oral board examinations, both individually and at a programmatic level, encounter similar difficulties such as lack of time (of learners and educators), financial constraints, and scarcity of standardized materials [18]. The challenges in implementing MOEs, despite their proven benefit, showcases the need to explore novel ways to leverage technology for delivering this education to trainees. By focusing on a custom LLM for surgical education, our study addresses a notable gap in evaluating AI's capability to manage the nuanced demands of examination preparation for residents, specifically oral board examination.

Limitations of the study

First, the resident recruitment methods may have introduced some self-selection bias among those who completed the transcripts. However, since we were not assessing user responses in this study, we believe this had minimal impact. Additionally, the small sample size of transcripts and surgical scenarios, along with evaluation by only two expert reviewers, necessitates caution when generalizing the model's effectiveness in surgical education and oral board preparation. Concerns for accuracy may stem from inherent limitations within the model itself. Trained on textual data from the internet, the model faced challenges due to the limited availability of publicly accessible surgical knowledge and specific information regarding general surgery oral board examinations.

Variability seen in the reviewers' evaluations and the lack of standardized metrics likely influenced the model's overall assessment and accuracy. We observed low inter-rater reliability, likely due to reviewers' differing opinions and biases about the model's capabilities and varying interpretations of the evaluation rubric. Despite standardizing instructions and providing iterative practice, Reviewer 1's early involvement in model and rubric development may have contributed to these discrepancies. Future studies should involve both reviewers in the iterative model-building process to mitigate biases and enhance their understanding of the model's capabilities. Given these limitations, further studies with an expanded pool of reviewers and surgical scenarios are necessary to comprehensively assess the model's efficacy before generalizing our results.

While human evaluation is commonly used to assess model output accuracy, additional metrics can be employed. Precision, recall, and F1 scores measure how closely the model's outputs match the correct answers or expected results [61]. Metrics like Bilingual Evaluation Understudy

(BLEU) and Recall-Oriented Understudy for Gisting Evaluation (ROGUE) compare generative text with reference text to assess accuracy [62]. However, we were unable to use similar metrics due to a lack of reference materials, as oral board topics and evaluation frameworks are not disclosed by the ABS, and the scarcity of publically available oral board scenarios. Without specific reference texts, accuracy measurement is challenging.

Lastly, the prompting strategy employed and the exclusive reliance on a commercial large LLM is a limitation of our study. The design and structure of prompts play a critical role in shaping the quality and relevance of the generated responses. Our study utilized a standardized prompting approach, which may not have fully optimized the model's potential performance. Exploring more sophisticated prompting techniques, such as ensembling and fine-tuning, could yield more accurate and contextually appropriate outputs. Furthermore, dependence on a single commercial LLM introduces constraints related to cost and potential biases inherent to the proprietary model. Utilizing a combination of open-source LLMs and ensemble methods could improve robustness, reproducibility, and allow for greater customization to specific educational needs. Future research should investigate these avenues to enhance the effectiveness and applicability of AI-driven tools in surgical education.

Future directions

Despite the challenges we faced, we discovered an unforeseen educational benefit. As our research team and reviewers engaged in critical analyses of the AI-generated content, it led to rich discussions and deeper insights into clinical reasoning and decision-making processes. This observation highlights the potential for AI tools to serve as an adjunct in the traditional education process, enhancing learning outcomes under expert guidance.

Looking ahead, the integration of domain-specific knowledge into these models presents a promising avenue for enhancing their educational efficacy by improving their accuracy. When considering the development of a more effective tool for simulating oral boards scenarios, it becomes evident that the quality of the tool is contingent upon the quality of content used to train the models. Therefore, initiatives such as the ABS's release of standardized materials and rubrics, alongside vetted mock oral scenarios and answers, could significantly enhance the accuracy and relevance of models like this.

This tailored approach to AI development ensures that the content not only addresses learners' educational needs, but also aligns with standards set by professional certification boards. Moreover, alongside incorporating domain specific knowledge, identifying errors in model-generated responses and leveraging surgical experts to refine future data sets will further enhance simulation accuracy. With large and accurate data sets, processes like RAG and fine-tuning can help create a highly precise, task-specific model.

As a community, with continual expansion of the technology, we must endeavor to establish a shared framework for evaluating these tools in the future. The traditional approaches to creating these frameworks, like the Delphi process, may prove ineffective given the accelerated and constant advancement of the technology. Therefore, we must innovate and develop streamlined evaluation methods that can keep pace with the rapid progression of AI technology. Once these AI models are thoroughly evaluated for independent use by trainees, future studies can compare their effectiveness to standard educational methods.

Conclusions

Our exploration into the use of generative AI as a board examiner during simulated oral scenarios reveals the potential of AI to transform traditional learning experiences in surgical education. We demonstrate that AI can generate and guide trainees through realistic clinical scenarios, though more work is needed to provide safe and accurate feedback about their performance. Our study underscores not only the potential to revolutionize educational experiences, but also the promise of transforming the traditionally time-consuming and costly resources required for them. Through collaborative efforts to refine these tools, guided by expert insights and standardized evaluation practices, we stand on the cusp of ushering in a new era of educational excellence. The integration of AI promises not only to enrich the educational landscape but also to democratize access to high-quality training resources, ultimately benefiting both learners and patients. This study contributes valuable insights into leveraging AI technology in surgical education, underscoring both its benefits and limitations, and paving the way for future advancements in this field.

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Data availability statement The chatbot prompts used in this study are available in the supplementary materials. Code used for statistical analysis in this study is available on GitHub [https://github.com/tsathe/surgbot-data-analysis]. Transcripts of the conversations between the users and the chatbot are available upon reasonable request, subject to review for participant privacy concerns. However, raw data from this study, including any potentially identifiable information, will not be shared to protect participant privacy.

Declarations

Conflict of interest Management: JR is a consultant for McGraw Hill, an American publishing company whose mission is the education of current and future healthcare professionals.

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